
Learning Conway’s Game of Life with CNN-Transformer Hybrids: Architecture Scaling, Exposure Bias, and Activation Inductive Bias

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 We study how a CNN-Transformer hybrid learns the one-step transition rule of
2 Conway’s Game of Life (GoL) on a 40×40 toroidal grid. Starting from a model
3 trained with pure teacher forcing, we show that autoregressive rollout suffers
4 from exposure bias, that a straight-through-estimator (STE) multi-step scheme
5 fails due to a systematic gradient bias, and that scheduled sampling resolves this
6 cleanly. We further show that once a model saturates its capacity, scaling the
7 training set five-fold *regresses* tail-case accuracy in a way invisible to aggregate
8 validation metrics, while scaling the model five-fold instead eliminates essentially
9 all remaining errors. Motivated by recent work on activation-function inductive
10 bias for cellular automata, we independently reproduce the finding that a learnable
11 second-degree polynomial activation lets a 25–34 parameter CNN learn GoL
12 reliably where ReLU fails, and report a negative result transferring this activation
13 to our larger multi-step model. Finally, we characterize the learned positional
14 embedding of our stable models: patch embeddings change measurably under
15 translation, well short of the CNN stage’s exact equivariance, but the change is
16 clearly separable from the models’ own baseline sensitivity to unrelated content —
17 and this separation grows, rather than shrinks, in our largest model.

18 1 Problem and architecture

19 GoL is a deterministic cellular automaton: each cell’s next state depends only on its own state and its
20 8-neighbor live count (B3/S23). We treat one GoL step as image-to-image prediction on a periodic
21 40×40 grid. Our core architecture is a CNN-Transformer hybrid: two circular-padded convolutions
22 extract local, translation-equivariant features respecting the torus topology; these are cut into non-
23 overlapping 4×4 patches, linearly projected, combined with a *learned* absolute positional embedding,
24 passed through a Transformer encoder for global mixing, and projected back to per-cell logits. GoL
25 is a natural stress test for learned representations: prior work shows plain feed-forward networks
26 struggle to learn it reliably at all [Springer and Kenyon, 2021], and the Transformer stage [Vaswani
27 et al., 2017] trades the CNN’s clean equivariance guarantee for long-range context (Section 3). This
28 V4 design is itself the fourth step of an earlier architecture search, each step fixing one identified
29 issue: V1 (avg-pooled tokens, learned positional embedding, zero-padding) $\rightarrow$ V2 (swaps in a *fixed*
30 *sinusoidal* positional embedding to remove a right-edge rollout artifact) $\rightarrow$ V3 (swaps avg-pooling
31 for a lossless flatten-and-project tokenization, reverting to a learned embedding) $\rightarrow$ V4 (swaps
32 zero-padding for circular padding, fixing incorrect boundary neighbor counts). V5 onward (Section 2)
33 all train this same V4 architecture under different regimes and sizes. V2’s checkpoint never stabilized

Ver.	Key change	Val. loss	Prec./Rec.	Status
V1	Baseline: avg-pool tokens, learned PE, zero-pad	— ^b	—	Edge artifacts; rollout freezes to a blob
V2	Learned PE → fixed sinusoidal PE	— ^b	—	Unstable training; no rollout improvement
V3	Avg-pool → flatten+linear tokenization	0.021	—	Fixes avg-pool bottleneck; first realistic rollouts
V4	Zero-padding → circular padding	0.226	100% / 94.3%	Fixes boundary bug; base arch. for V5+
V5	Single-step teacher forcing	0.201	99.4% / 99.7%	Collapses under rollout
V6	+STE multi-step	0.234 ^a	92.7% / —	Diverges
V7	STE → scheduled sampling	0.201	99.7% / 100%	Stable
V8	+3× high-density upsample	0.201	100% / 99.9%	Best per-grid (291K params)
V9	+5× all data	0.201	99.7% / 99.6%	Regressed tail errors
V10	5.15× model size	0.199	100% / 100%	Zero errors , 2,500/2,500

Table 1: Version summary, spanning the V1–V4 architecture search (Section 1) and the V5–V10 training-regime search (Section 2). ^aV6’s best epoch before divergence. ^bV1/V2 reported only aggregate val accuracy (89.4% and 84% respectively), not loss or a precision/recall split.

(Table 1), so it is not used for any further analysis; Section 3 instead characterizes the learned positional embedding using only the stable, well-converged V8 and V10 models.

2 Training evolution: two failure modes and their fixes

Exposure bias. A model trained with pure single-step teacher forcing (V5) scores > 99% precision/recall on held-out single steps but collapses under autoregressive rollout, since it never sees its own (imperfect) outputs during training. A straight-through-estimator (STE) multi-step scheme intended to fix this instead **diverges** after epoch 5, which we trace to a systematic gradient bias from the STE’s hard threshold. Replacing STE with *scheduled sampling* [Bengio et al., 2015] — gradually increasing the probability of feeding the model its own previous prediction — fixes this cleanly and stably (V7).

A hidden data-scaling regression. Upsampling high-density configurations 3× (V8) improves per-grid accuracy further (291K params). Naively scaling the *training set* 5× (V9) leaves aggregate validation metrics statistically unchanged, but a density-stratified failure search (500 grids at each of five densities, fixed seed across models) reveals V9’s error count at density 0.80 is nearly **seven times worse** than V8’s (22,190 vs. 3,358 summed false positives/negatives) — invisible in aggregate metrics alone. Instead of more data, we scale the *model*: V10 (5.15× V8’s size, 1.5M params, trained from scratch) achieves **zero errors** across all 2,500 test grids at every density (Table 1, Figure 1). The lesson: once capacity saturates, more data can silently trade average-case for tail-case performance, detectable only by per-subgroup error auditing, not validation loss.

3 Embedding geometry: how equivariant is a learned positional embedding?

A CNN feature map is a *physical* embedding: convolution’s translation-equivariance means shifting the input shifts the feature map identically. A Transformer’s patch embeddings, after a learned absolute positional embedding, have no such guarantee — each position slot is an independently learned vector unrelated to its neighbors. An earlier version of this analysis tried to test this by comparing V8 against V2, a fixed-sinusoidal-PE variant whose angle-addition identities give closed-form near-equivariant structure. We dropped that comparison: V2’s checkpoint is from epoch 1 of a run that never stabilized afterward (Table 1), and a follow-up check showed V2 could barely distinguish two *different* random grids of different density (cosine similarity 0.958) — almost as similar as it rated the *same* grid before and after translation (0.997). An apparent equivariance advantage that shows up most strongly on exactly the inputs a model cannot tell apart is evidence of an undertrained, indiscriminative embedding, not of sinusoidal positional structure, so we do not report it as a finding.

Instead we ask a narrower, defensible question using only stable models: how much does V8’s (and V10’s) learned positional embedding change under translation, relative to each model’s *own*

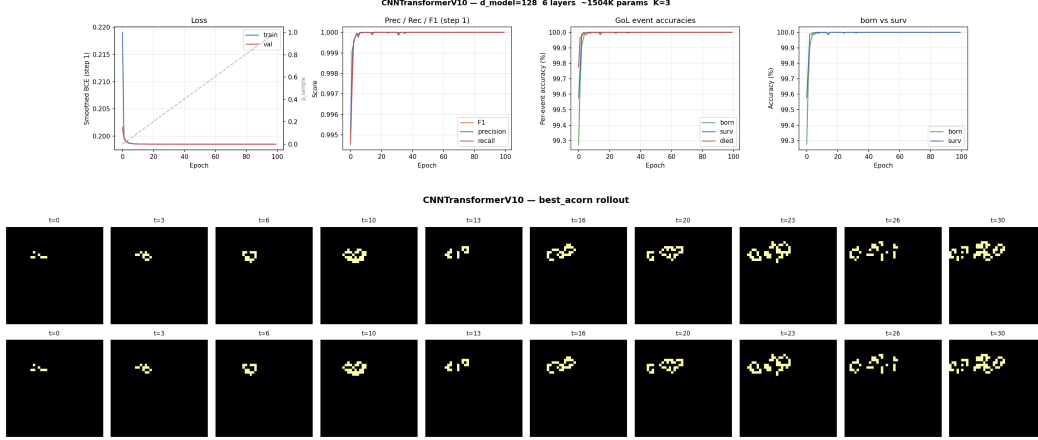


Figure 1: Top: V10 training curve. Bottom: a 30-step autoregressive rollout of the *acorn* methuselah — a 7-cell seed that grows into a large, long-lived chaotic pattern — true trajectory (top row) vs. V10’s autoregressive prediction (bottom row); V10 tracks the true trajectory exactly at every shown step, and makes zero errors across 2,500 held-out grids overall.

baseline sensitivity to unrelated content? For three configurations (glider, and two random grids at densities 0.35 and 0.65) we measure post-Transformer cosine similarity between (i) a grid’s embedding and a translated copy’s embedding (“translation similarity”), and (ii) embeddings of the different, untranslated configurations against each other (“content baseline”). A real, non-trivial equivariance signal should show (i) clearly above (ii), without (ii) itself being suspiciously close to 1.0 (collapse). V8 shows translation similarity 0.890 against a content baseline of 0.803 (gap +0.087); V10 shows translation similarity 0.690 against a content baseline of 0.494 (gap +0.197) (Figure 2, top). Both models are well short of exact equivariance (similarity 1.0), but both show a translation-specific signal clearly separated from their own content-discrimination baseline — and that separation is over twice as large in V10, our larger, more capable model, even though V10 is also more discriminative overall (its lower content baseline). Scaling model capacity did not erode this partial equivariance signal; if anything it grew alongside the model’s general discriminative power.

4 Independent verification: polynomial activations for GoL

Motivated by a recent claim that a learnable second-degree polynomial activation $\phi(x) = w_0 + w_1x + w_2x^2$ lets a minimal CNN learn GoL reliably where ReLU fails almost completely [Ahmed and Davis, 2026], we independently reproduced this rather than taking it on faith. We built two CNNs matching the paper’s minimal architecture (two 1×1 convolutions plus one 3×3 circular-padded convolution), **identical except for the activation function**, matching the paper’s reported parameter counts exactly (ReLU: 25 params, polynomial: 34 params). Trained single-step, teacher-forcing, over 10 random seeds: ReLU converges to perfect validation accuracy in only 1/10 seeds (mean final accuracy 89.4%), while the polynomial activation converges in 10/10 seeds (100.0% mean accuracy), confirming the claim on our own toroidal GoL setup (Figure 2, bottom). We then attempted to transfer this activation into our larger multi-step model (same size as V8) by replacing GELU with polynomial activations in the CNN stage; this **failed**, diverging sharply after epoch 5 under scheduled sampling, likely because the unbounded w_2x^2 term compounds through the autoregressive feedback loop in a way it cannot in a single-step setting. Candidate fixes (a bounded/saturating variant, a separate learning rate for polynomial coefficients) were identified but not yet pursued.

5 Conclusion

Three practical lessons emerge. First, autoregressive exposure bias is best fixed with scheduled sampling rather than a hard differentiable relaxation (STE), which can introduce its own destabilizing gradient bias. Second, aggregate validation metrics can hide severe, density- specific regressions;

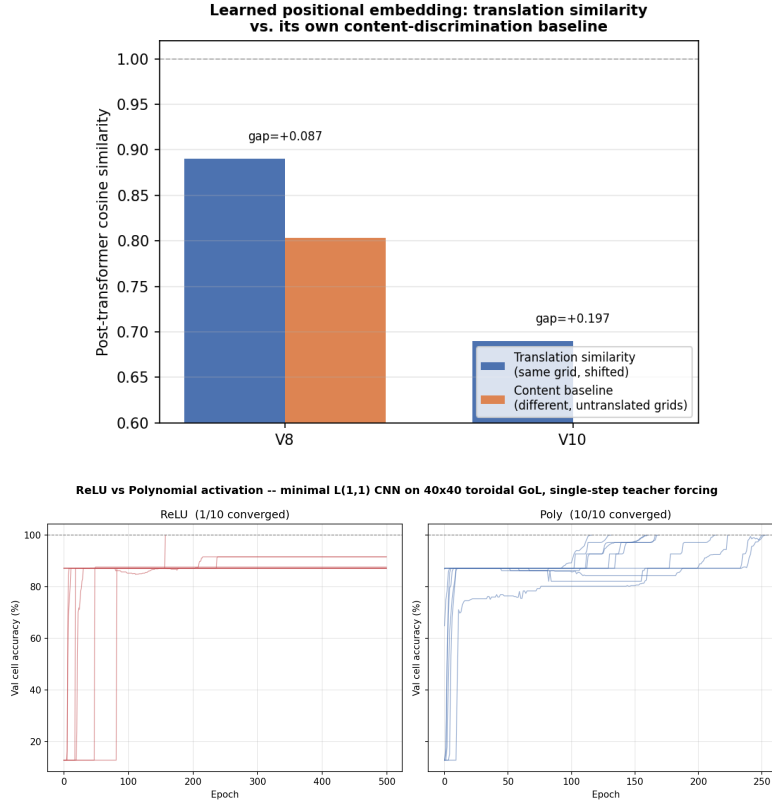


Figure 2: Top: for each stable model (V8, V10), post-Transformer cosine similarity under translation (“translation similarity”) vs. between different untranslated configurations (“content baseline”); the gap between the two bars is the model’s translation-specific signal, net of its own baseline sensitivity to content. Bottom: validation accuracy over training, 10 seeds per activation, on a minimal 25/34-parameter CNN; polynomial activation converges reliably where ReLU does not.

per-subgroup error auditing is necessary once a model is scaled with more data rather than more capacity. Third, an activation function’s inductive-bias advantage demonstrated at minimal scale and single-step training does not automatically transfer to a larger, multi-step, autoregressively-trained model — the interaction between an activation’s stability properties and the training regime matters as much as the activation itself.

References

- Tashin Ahmed and Q. Tyrell Davis. It’s much easier for neural networks to learn game of life dynamics with the right activation function: Polynomial kolmogorov-arnold networks. *arXiv preprint arXiv:2606.23587*, 2026.
- Samy Bengio, Oriol Vinyals, Navdeep Jaitly, and Noam Shazeer. Scheduled sampling for sequence prediction with recurrent neural networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2015.
- Jacob M. Springer and Garrett T. Kenyon. It’s hard for neural networks to learn the game of life. *International Joint Conference on Neural Networks (IJCNN)*, 2021.
- Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.